

# HATM — Human and AI Trust Compatibility Module

**Parent Standard:** Trust Layer Standard

**Category:** Governance & Enforcement

**Subcategory:** Human and AI Trust Compatibility

**Type:** Trust Architecture Module

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**Compatibility:** OOF Methodology OS · Trust Layer Standard · Truth Validation Layer · Integrity Standard · Universal Canonical Language · Origin-Bound Identity · Audit in Real Time · AI Runtime Integrity Vault

**Authority:** OOF

**Protection:** MIP — Methodological Intellectual Property

**Canonical Language:** English

## Canonical Definition

**Human and AI Trust Compatibility Module** defines the structural conditions under which trust may remain compatible, understandable, and continuously verifiable across both human interpretation and AI-executed operational logic.

A system satisfies HATM only if:

- trust conditions remain readable for humans and machine-operable for AI systems
- trust does not depend on human-only assumptions or machine-only logic
- meaning remains stable across human and AI interpretation
- operational trust can be verified without contradiction between human judgment and machine execution
- trust compatibility remains structurally preserved across hybrid environments

A system that appears trustworthy to humans but cannot be operationally validated by AI, or appears valid to AI while remaining unintelligible or misleading to humans, does not satisfy HATM.

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## Module Function

HATM defines the compatibility layer of trust architecture between human and machine-governed environments.

It ensures that trust is not split into two disconnected realities:

- one for human interpretation
- one for machine execution

The module applies wherever trust must remain stable across:

- human oversight
- AI-supported systems

- autonomous execution
- hybrid operational environments
- cross-system interpretation
- machine-readable governance
- human-facing validation conditions

Its function is to preserve one structurally compatible trust-state across both human and AI-operated environments.

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## Step 1 — Define the Trust Compatibility Object

The organization must define what trust relationship must remain compatible across humans and AI systems.

Minimum requirement:

- the trust object is explicit
- the compatibility scope is bounded
- undefined compatibility targets are excluded from valid trust logic

The trust object may include a:

- system
  - process
  - workflow
  - AI agent
  - operational decision path
  - hybrid human-AI environment
  - machine-assisted governance structure
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## Step 2 — Define Human Trust Conditions

The system must define what makes the trust-state understandable and valid from the human side.

Minimum requirement:

- human trust conditions are explicit
- the trust-state is interpretable by accountable human actors
- trust does not depend on opaque machine output alone

These conditions may include:

- understandable meaning
  - accountable decision logic
  - reviewable validation status
  - visible responsibility structure
  - non-deceptive interpretation boundaries
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## Step 3 — Define AI Trust Conditions

The system must define what makes the trust-state machine-readable and operationally valid from the AI side.

Minimum requirement:

- AI trust conditions are explicit
- trust logic is interpretable under machine-operable structural rules
- trust does not depend on vague human intuition alone

These conditions may include:

- canonical meaning consistency
  - machine-readable validation logic
  - runtime traceability
  - structured integrity conditions
  - bounded execution rules
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## Step 4 — Define Compatibility Logic

The system must define how human trust conditions and AI trust conditions remain aligned.

Minimum requirement:

- compatibility logic is explicit
- contradiction between human-readable trust and machine-executed trust can be identified
- trust compatibility is not assumed merely because both humans and AI participate

This means the system must ensure that:

- what humans believe they are trusting
  - and what AI is actually executing under trust conditions
  - remain structurally compatible.
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## Step 5 — Define Meaning Stability Across Human and AI Interpretation

The system must define how trust-related meaning remains stable across both human and machine interpretation.

Minimum requirement:

- trust-related terms remain canonically consistent
- semantic drift between human reading and AI execution is restricted
- trust compatibility does not collapse through divergent interpretation of the same condition

Without stable meaning, compatibility becomes symbolic rather than operational.

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## Step 6 — Preserve Compatibility Traceability

The system must preserve traceability of human and AI trust alignment.

Minimum requirement:

- compatibility evidence is reviewable
- trust alignment remains auditable
- later reconstruction of trust compatibility remains possible

If compatibility cannot be reconstructed, then trust across humans and AI becomes another unverified claim.

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## Step 7 — Restrict Invalid Human and AI Trust Design

The system must not be treated as valid if trust appears aligned only at the surface while structural incompatibility remains underneath.

Minimum requirement:

- invalid trust compatibility conditions are identifiable
  - symbolic alignment is excluded
  - trust cannot be claimed as compatible if humans and AI operate under different effective trust logic
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## Scenario

A human supervisory team relies on an AI-supported system for operational decisions, but the machine interprets trust conditions differently from the human reviewers.

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## Application

HATM is used to ensure:

- trust conditions are explicit for both human and AI interpretation
  - canonical meaning remains stable across both sides
  - compatibility gaps between human oversight and machine execution can be identified
  - trust does not appear aligned only because both sides use the same words differently
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## Result

The organization gains a stronger trust architecture in which human and AI trust conditions remain structurally compatible rather than only superficially similar.

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## Scenario

A hybrid environment combines human operators, automated systems, and AI execution layers across a shared workflow where trust must

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remain stable across all participants.

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## Application

HATM is used to ensure:

- trust remains readable for humans and executable for machines
  - compatibility is preserved across interpretation and operation
  - trust-state does not fragment between human assumptions and machine logic
  - structural contradictions can be detected before operational failure occurs
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## Result

The environment gains a stronger basis for trust continuity because human and AI participation remain aligned within one compatible trust structure.

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## Canonical Closing Statement

**If humans and AI cannot trust under the same structural conditions, trust becomes fragmented before the system fails.**

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## Related Documents

→ [Trust Layer Standard \(TLS™\)](#)

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## Canonical Source

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